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Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

ITEP 414 – System Administration and Maintenance

Week 2 Portfolio Project

Enterprise Infrastructure Planning for a Startup Company

Course: ITEP 414 – System Administration and Maintenance

Program: Bachelor of Science in Information Technology

Week: 2

Project Type: Individual Portfolio Project

Points: 100

Submission Platform:

- GitHub Repository
- LMS
- LinkedIn

Project Overview

Every successful IT infrastructure begins with proper planning.

Before purchasing computers, installing servers, configuring networks, or deploying cloud services, a System Administrator must first understand the organization's business requirements and design an infrastructure that supports business operations.

In this project, you will assume the role of a **Junior System Administrator** assigned to prepare the initial IT Infrastructure Plan of a newly established startup company.

Your final output should resemble a professional document that could be submitted to an IT Manager or Company Executive.

Learning Objectives

At the end of this project, students should be able to:

Knowledge

- Explain the roles and responsibilities of a System Administrator.
- Identify the hardware, software, and networking requirements of a small business.
- Describe the purpose of IT documentation and infrastructure planning.

Skills

Students will be able to:

- Analyze organizational IT requirements.
- Prepare professional IT inventories.
- Design an enterprise network topology.
- Create technical documentation using Markdown.
- Present infrastructure planning professionally.

Attitude

Students are expected to demonstrate:

- Professionalism
- Organization
- Technical communication
- Attention to detail
- Critical thinking

Prepared by: **JOHN RANDOLF M. PENAREDONDO, MIT**

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Project Scenario

You have recently been hired as the **Junior System Administrator** of **ABC Startup Solutions**, a newly established software development company.

The company currently has **20 employees** and occupies a single office floor. Management has requested that you prepare the complete IT Infrastructure Plan before purchasing equipment.

The company consists of the following departments:

Department	Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
TOTAL	20

The company currently has:

- No computers
- No server
- No network
- No internet infrastructure
- No security policies

Your task is to design everything from scratch.

Project Instructions

Your final project must contain **eight sections**.

PART 1

Company Profile

Create a company profile.

Include:

- Company Name
- Nature of Business
- Company Vision
- Company Mission
- Office Location (fictional)
- Organizational Structure
- Employee Distribution

PART 2

Enterprise Hardware Inventory

Prepare a professional hardware inventory.

Minimum required hardware:

- Desktop Computers
- Laptops
- Server
- Router

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- Switch
- Printer
- UPS
- Wireless Access Point
- NAS Storage
- External Backup Drive
- Monitors

Create a table similar to this:

Asset ID Hardware Quantity Department Purpose

Provide realistic quantities and justify your selections.

PART 3

Enterprise Software Inventory

Prepare the software inventory.

Include at least:

- Windows 11 Pro
- Ubuntu Server
- Microsoft Office
- VS Code
- Git
- GitHub Desktop
- VirtualBox
- Google Chrome
- Microsoft Defender
- AnyDesk
- 7-Zip

Table Format

| Software | Version | License | Purpose |

Explain why each software is needed.

PART 4

Enterprise Network Inventory

Prepare the networking equipment inventory.

Include:

- ISP Modem
- Router
- Firewall
- Switch
- Access Point
- Patch Panel
- CAT6 Cables
- RJ45 Connectors

Create a professional inventory table.

PART 5

Enterprise Network Diagram

Using **Draw.io**, design a professional network topology.

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Your diagram must include:

- Internet
- ISP Modem
- Router
- Firewall
- Switch
- Wireless Access Point
- Server
- Printer
- IT Department
- HR Department
- Finance Department
- Sales Department

Requirements:

- Proper networking symbols
- Logical connections
- Labels
- Diagram Title
- Student Name

Export as:

- PNG
- PDF

PART 6

System Administration Roles

Research and explain the following positions.

Helpdesk Technician

Discuss:

- Responsibilities
- Skills
- Common Tools
- Certifications

Network Administrator

Discuss:

- Responsibilities
- Skills
- Common Tools
- Certifications

Linux System Administrator

Discuss:

- Responsibilities
- Skills
- Common Tools
- Certifications



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Cloud Administrator

Discuss:

- Responsibilities
- Skills
- Common Tools
- Certifications

Finally,

Write one paragraph explaining **how these four professionals work together inside one organization.**

PART 7

Infrastructure Recommendations

Write your recommendations.

Discuss:

- Internet Provider
- Server Specifications
- Backup Strategy
- Security Recommendations
- Antivirus
- Password Policy
- Expansion Plan

Support your recommendations with logical reasons.

PART 8

Personal Reflection

Write a **300-500 word reflection** discussing:

- What did you learn?
- Which task was most challenging?
- Why is planning important before deployment?
- How will this project help you become a better System Administrator?

Documentation Requirements

Inside your GitHub repository:

BSIT-SystemAdministration-Portfolio/

```
|
|— Week02/
|   |— EnterpriseInfrastructurePlan.pdf
|   |— README.md
|   |— diagrams/
|   |— images/
|   |— references/
```

README.md Format

Your README should include:

- Project Overview
- Learning Objectives
- Company Scenario

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- Hardware Inventory Summary
- Software Inventory Summary
- Embedded Network Diagram
- Technologies Used
- Challenges Encountered
- Reflection
- References

Use Markdown headings, tables, images, and hyperlinks to make the README professional and easy to navigate.

LinkedIn Portfolio Task

Publish a professional LinkedIn post summarizing your Week 2 project.

Your post should include:

- A brief description of the startup company scenario.
- Key skills practiced (planning, inventory, documentation, network design).
- One screenshot of your network diagram or inventory.
- A link to your GitHub repository.
- Relevant hashtags such as **#SystemAdministration**
#InfrastructurePlanning **#Networking** **#GitHub** **#BSIT** **#LearningInPublic**.

Submission Checklist

Before submitting, verify that you have completed the following:

Requirement	Completed
Company Profile	☐
Hardware Inventory	☐
Software Inventory	☐
Network Inventory	☐
Network Diagram (PNG & PDF)	☐
IT Roles Research	☐
Infrastructure Recommendations	☐
Reflection	☐
README.md	☐
GitHub Repository Updated	☐
LinkedIn Post Published	☐



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Grading Rubric (100 Points)

Criteria	Excellent (A)	Good (B)	Satisfactory (C)	Needs Improvement (D/F)	Points
Business Analysis & Company Profile	Company profile is complete, realistic, and professionally presented with clear organizational structure.	Minor details missing but mostly complete.	Basic profile provided with limited detail.	Incomplete or unrealistic profile.	10
Hardware, Software & Network Inventory	Inventories are comprehensive, accurate, justified, and formatted professionally.	Mostly complete with minor omissions or formatting issues.	Some required items missing or weak justification.	Inventories are incomplete or inaccurate.	20
Enterprise Network Diagram	Diagram is accurate, well-organized, properly labeled, uses standard networking symbols, and clearly reflects the scenario.	Minor layout or labeling issues.	Functional but lacks clarity or contains several errors.	Diagram is incomplete, confusing, or missing major components.	20
IT Roles Research & Analysis	Demonstrates strong understanding of all four roles, supported by reliable sources and a clear explanation of collaboration.	Good explanations with minor missing details.	Basic descriptions with limited analysis or weak sources.	Research is incomplete, inaccurate, or lacks citations.	15
Infrastructure Recommendations	Recommendations are practical, technically	Mostly appropriate with limited	General recommendations with	Recommendations are unrealistic,	10

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	sound, justified, and aligned with the company's needs.	justification.	minimal reasoning.	unsupported, or incomplete.	
Technical Documentation (README & Report)	Documentation is complete, professionally written, well-organized, visually appealing, and free from major grammar or formatting issues.	Minor writing or formatting issues.	Documentation is understandable but lacks organization or polish.	Documentation is incomplete or poorly presented.	15
GitHub Repository & Professional Submission	Repository is organized, follows the required structure, includes all deliverables, and demonstrates professional version control practices. LinkedIn post is complete and professional.	Minor repository organization issues or minor omissions in the LinkedIn post.	Repository is functional but lacks organization or required elements.	Repository is incomplete, inaccessible, or submission requirements are not met.	10
				TOTAL	100 Points

Expected Portfolio Artifact

By completing this project, you will add your **second professional portfolio artifact** to your GitHub repository: an **Enterprise Infrastructure Planning Project**. This project demonstrates your ability to analyze business requirements, design foundational IT infrastructure, document technical decisions, and communicate professionally—competencies expected of an entry-level System Administrator and valued by employers during internship and job applications.